

Satyaranjan Biswal

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Professional Experience

Marie Skłodowska-Curie Actions (MSCA) Fellow *Oct 2020 – Present*

Fritz Haber Research Centre for Molecular Dynamics, Hebrew University of Jerusalem

- **Advisor:** Prof. Noam Agmon

- **Research:** Proton transport in biological systems using classical and quantum molecular dynamics.

PhD Secondment *Dec 2021 – Mar 2022*

Freie Universität Berlin, Germany

- Supervised by Prof. Noam Agmon and Prof. Ana-Nicoleta Bondar.

- Developed carboxylate-enriched collagen peptides as proton-conducting "wires" for molecular proton transport.

Research Scholar *Oct 2017 – Aug 2020*

Department of Chemistry, IISER Berhampur, India

- **Advisor:** Prof. M.K Rana

- Developed molecular simulations and force fields.

- Investigated anion sensing mechanisms in drug delivery.

Dissertation Research *Feb 2017 – Sep 2017*

Molecular Biophysics Unit, IISc Bangalore, India

- **Advisor:** Prof. Manju Bansal F.N.A., F.N.A.Sc., F.A.Sc

- Genome-wide structural analysis of 5'UTR introns in *Oryza sativa*.

Machine Learning-Based Virtual Screening (Collaborative Project) *2024 – Present*

Division of Agricultural Bioinformatics, ICAR-IASRI, New Delhi, India

- Developed an AI-driven web server for predicting protein-ligand affinities validated via MD simulations.

Education

Ph.D. in Chemistry *Oct 2020 – Present*

Hebrew University of Jerusalem, Israel

- **Thesis:** Proton and Ion Movement on Hydrated Tropocollagen Surfaces

- **Advisor:** Prof. Noam Agmon

M.Sc. in Bioinformatics *2015 – 2017*

Orissa University of Agriculture and Technology, India

B.Sc. in Biological Science *2012 – 2015*

Utkal University, India (with Honors)

PhD Thesis Publications

- Biswal, S., Agmon, N. *Collagen structured hydration*. Biomolecules, 2023.
- Biswal, S., Agmon, N. *Cation Conduction along Collagen Wires*. (Submitted to Science Advances)
- Biswal, S., Agmon, N. *Proton Movement Along Hydrated Collagen Surfaces*. (In preparation)

Other Peer-Reviewed Publications

- Mahanandia, N.C., Biswal, S., et al. *NatProCP: A Plant-Derived Natural Products Database for Crop Protection Targeting SDH1 Protein of Magnaporthe oryzae*. Biochem. Biophys. Res. Commun., 2026. (Accepted)
- Mahanandia, N.C., Biswal, S., et al. *Bioinformatics Tools: Insights from Structural Approaches*. Adv. Omics Technol., 2025, 177–191.
- Mahanandia, N.C., et al. *Rice blast disease immunity targeting CPKA protein*. J. Biomol. Struct. Dyn., 2024.
- Panda, S.K., et al. *Allosteric binding site in SARS-CoV-2 papain-like protease*. J. Phys. Chem. Lett., 2023.
- Biswal, S., et al. *Binding mechanism of ascorbic acid and violaxanthin with violaxanthin de-epoxidase enzymes*. Photosynth. Res., 2023.
- Srivastava, R., et al. *Exploring antiviral potency of N-1 substituted pyrimidines against HIV-1 and other DNA/RNA viruses*. Comput. Biol. Chem., 2023.
- Sen Gupta, P.S., Biswal, S., et al. *Binding insights into COVID-19 protein target with ivermectin*. J. Biomol. Struct. Dyn., 2022.
- Das, N.C., et al. *Filarial cystatin ligand of human TLR4*. J. Biomol. Struct. Dyn., 2022.
- Ray, A.K., et al. *FDA drugs repurposing against SARS-CoV-2 main protease*. Comput. Biol. Med., 2022.
- Panda, S., Biswal, S., et al. *ACE-2-derived biomimetic peptides for inhibition of SARS-CoV-2 spike protein*. J. Proteome Res., 2021.
- Bag, S., et al. *Rec-BSP1 as a fertility factor in buffalo*. bioRxiv, 2021.
- Sen Gupta, P.S., Biswal, S., et al. *Binding insight of clinically oriented drug famotidine with the identified potential target of SARS-CoV-2*. J. Biomol. Struct. Dyn., 2021.
- Gupta, P.S.S., et al. *Computer-aided discovery of bis-indole derivatives as multi-target drugs*. J. Mol. Liq., 2020.
- Bhat, H.R., et al. *Anion sensing by novel triarylboranes*. ACS Omega, 2019.

Awards & Honors

- Best Poster Award, TCS Conference, IIT Madras (2023)
- DADA Fellowship, RWACH Achen, 2020
- DRG Fellowship, Leipzig University, 2020
- Dean's List (S. Swaminathan Student Award), 2017

Technical Skills

- **Computational Chemistry:** GROMACS, AMBER, NAMD, Gaussian, CP2K, ORCA, VASP, DFTB+.
- **Programming:** Python, C, C++, Fortran, Tcl, Shell scripting.
- **Machine Learning:** Scikit-learn, TensorFlow, Keras, XGBoost.
- **Visualization Tools:** VMD, Chimera, ChimeraX, PyMol.
- **High-Performance Computing:** HPC cluster simulations.

References

- **Prof. Noam Agmon**
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- **Prof. Malaya Kumar Rana**
Associate Professor, IISER Berhampur
Email: mrana@iiserbpr.ac.in
- **Dr. Parth Sarthi Sen Gupta**
Associate Professor, D Y Patil International University & IISER Berhampur
Email: parth.gupta@dypiu.ac.in